NS-3 MANET ROUTING EXPERIMENT

Adaptive DSR Route Cache

Parameter sweep comparing mobility-scaled LinkCache timers against fixed DSR cache behavior

Thesis

In high mobility, DSR should trust cached routes for less time. The experiment tests whether speed-scaled cache timers improve delivery.

50
NODES
Random waypoint MANET

10
FLOWS
one-to-one UDP pairs

4
SWEEP POINTS
10%, 50%, 110%, 150%

200s
RUNTIME
one CSV row per second

PROBLEM

DSR cache reuse is valuable until mobility turns cached routes stale.

Why the cache can hurt

- DSR avoids repeated route discovery by keeping learned source routes and link state.
- In RandomWaypoint mobility, a route that worked earlier can point through a now-broken link.
- Long cache lifetimes can keep stale state alive long enough to drop packets before repair.

Adapt cache trust to movement

The experiment keeps the default MANET scenario and metrics, but computes cache timers from node speed before installing DSR.

Higher node speed -> shorter route and link stability timers

FORMULA

A single mobility scale converts base cache timers into effective DSR values.

```
adaptive_value = clamp(base_value * cache_reference_speed / node_speed,
min_value, max_value)
```

Formula variables

base_value: original timer multiplied by the sweep percentage
cache_reference_speed: speed where base timers are preserved, 5 m/s
node_speed: scenario speed used for cache trust, 20 m/s
clamp: keeps timers inside configured safety bounds

This sweep used a 0.25 mobility scale

cache_reference_speed / node_speed = 5 / 20 = 0.25

Example: the 50% timeout starts at 150 s, then becomes clamp(150 * 0.25, 30, 300) = 37.5 s.

30-300s
ROUTECACHETIMEOUT
route lifetime bounds

2-25s
INITSTABILITY
new link confidence

10-120s
USEEXTENDS
successful-use extension

IMPLEMENTATION

The adaptive values are computed before DSR is installed on the nodes.

Command-line switches

adaptiveRouteCache
adaptiveLinkStability
cacheType
baseRouteCacheTimeout
baseInitStability
baseUseExtends
cacheReferenceSpeed

```
Config::SetDefault("ns3::dsr::DsrRouting::Route
CacheTimeout",

TimeValue(Seconds(m_effectiveRouteCacheTimeout)
));
Config::SetDefault("ns3::dsr::DsrRouting::InitS
tability",

TimeValue(Seconds(m_effectiveInitStability)));
Config::SetDefault("ns3::dsr::DsrRouting::UseEx
tends",

TimeValue(Seconds(m_effectiveUseExtends)));
```

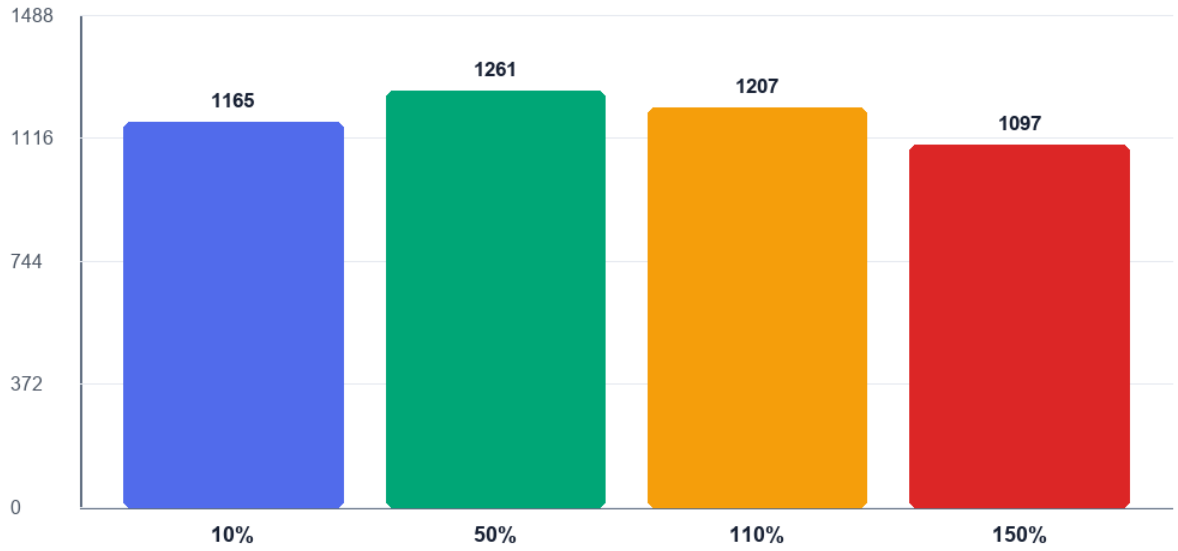
Placement matters: defaults must be set before dsrMain.Install(...) creates the routing objects.

ADAPTIVE SWEEP

The 50% adaptive setting delivered the strongest packet count in this run.

Adaptive sweep: total packets

Packets received over 200 seconds



1261

BEST PACKET COUNT

50% adaptive run

3.228

AVG KBPS

full 200-second average

37.5s

EFFECTIVE TIMEOUT

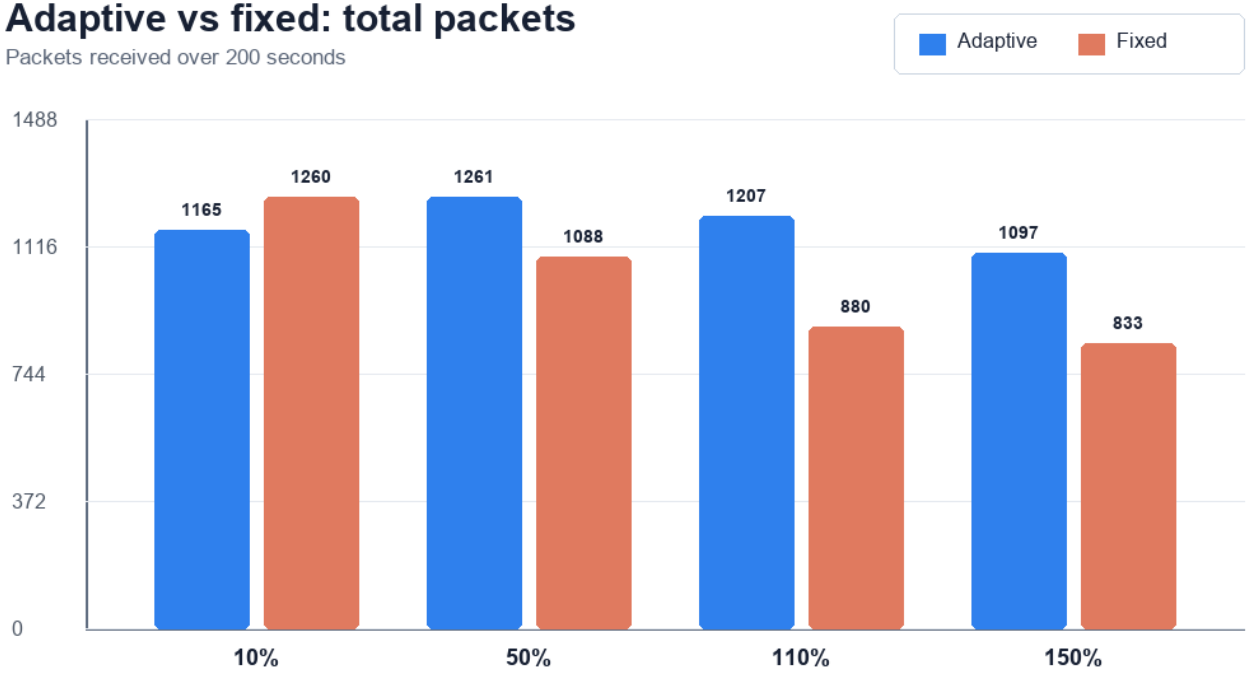
50% base after scaling

FIXED COMPARISON

Adaptive tuning beat fixed cache behavior at 50%, 110%, and 150%.

Adaptive vs fixed: total packets

Packets received over 200 seconds



+173

50% PACKET DELTA

adaptive vs fixed

+327

110% PACKET DELTA

largest improvement

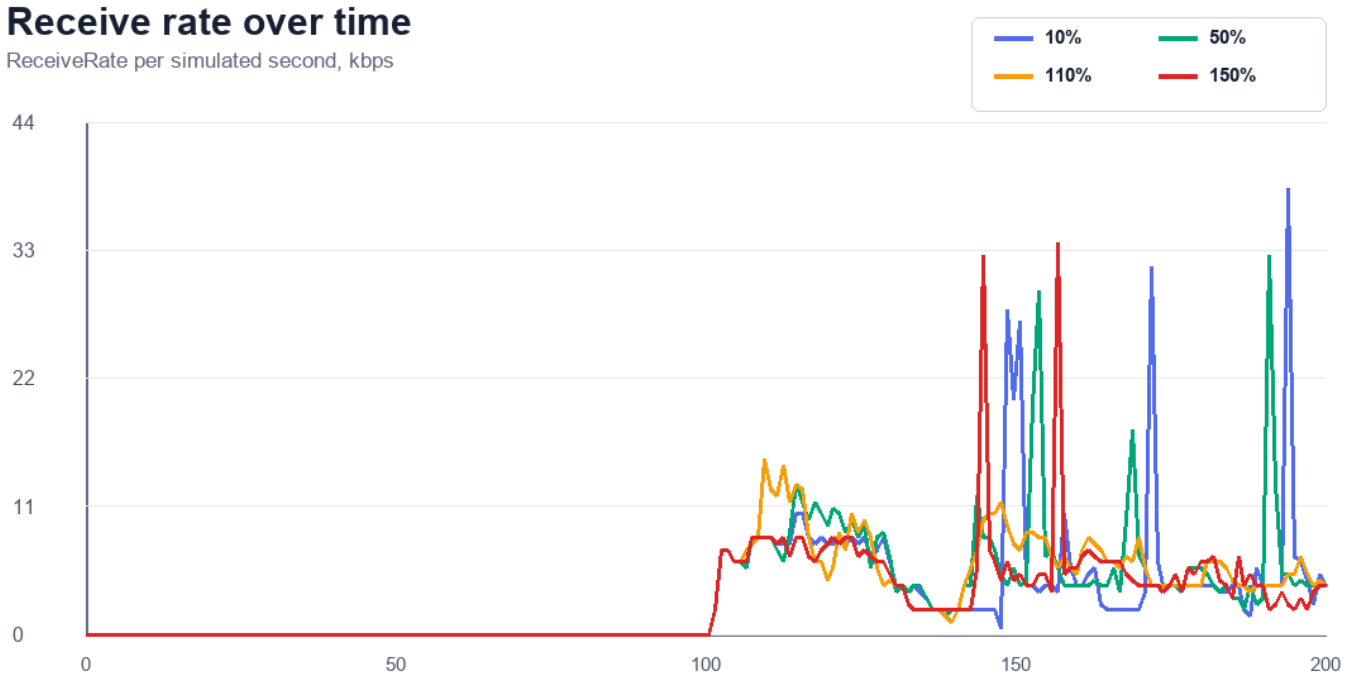
-95

10% PACKET DELTA

low-end clamp loses

TIME SERIES

Packet delivery starts after traffic begins and differs mainly during active seconds.



How to read it

Rows are one simulated second. Traffic starts around 100 s, so the first half of each metrics file is intentionally zero. The comparisons use total packets and average kbps across the full 200 s for consistency with the default ns-3 example.

TAKEAWAYS

Adaptive cache timing looks promising, but it needs seed and scenario sweeps next.

What the experiment suggests

- A moderate effective timeout outperformed both very short and long-lived cache assumptions.
- The adaptive policy helped most when fixed DSR would otherwise keep stale cache state too long.
- The 10% case exposes the impact of clamp bounds, so low-end tuning deserves separate attention.

Future work

Repeat runs across random seeds
Sweep cacheReferenceSpeed independently
Compare LinkCache and PathCache
Sweep node speed and node density
Track routing overhead alongside delivery

3/4
ADAPTIVE WINS
against fixed comparator

50%
BEST SWEEP POINT
1261 packets received

next
VARIANCE TESTING
multi-seed validation